Preclass Learning: Interactive Visualization (ICA 18)

1. Core Interactivity Features

Interactivity provides the user with control over the visualization, moving beyond a passive experience.

- **Filtering and Subsetting:** Allowing users to select specific categories or ranges to display (e.g., using a dropdown menu or slider).
- **Tooltips (Details on Demand):** Revealing specific data values when a cursor hovers over a data point.
- **Zooming and Panning:** Enabling a user to inspect dense areas of the plot or change the view.
- **Linking and Brushing:** Highlighting corresponding elements across multiple visualizations simultaneously when one is selected.

2. Interactive Frameworks

- **R/Python Libraries (Plotly, Leaflet):** Packages that wrap powerful JavaScript libraries, allowing data scientists to create web-ready interactive graphics directly from their code.
- **Web Applications (Shiny/Dash):** Frameworks that allow the creation of full data apps where changes in user input (widgets) automatically trigger reactive updates in the data processing and visualization layers.

3. Design and Performance

- **Intuitive Controls:** Interactive widgets must be clearly labeled and logically grouped to guide user exploration.
- **Performance Optimization:** Since the visualization redraws based on user input, the underlying data queries must be fast. Data often needs to be pre-aggregated or structured for rapid filtering.
- **Responsive Design:** Ensuring the controls and the plot scale correctly and remain usable on various devices (desktop, tablet, mobile).